Group D: Late progression

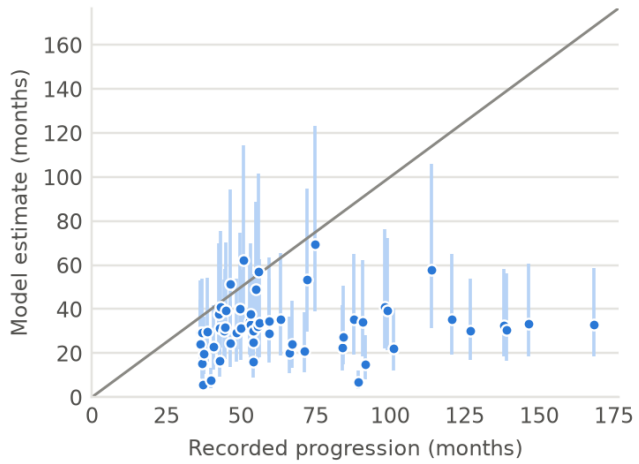
PrimeKG Patient Reasoning · Precision Oncology Research

Group D · 56 patients

Report date 14 Oct 2026

Group D · Late progression recorded progression-free survival: > 36 months

Patients	Median model risk percentile	Recorded progression	Model window agreement
56	**51**	**54.5 mo**	**25%**
5.2% of the cohort	cohort-wide risk ranking, out-of-fold	median; range 36.2–168.2 mo	estimate in this window · median |error| 23 mo



Most frequent path end-points in this group	Patients
SUMOylation of DNA damage response and repair proteins	6
Estrogen-dependent gene expression	5
Synthesis of PA	4
Extra-nuclear estrogen signaling	4
Post-translational protein phosphorylation	4
Interaction between L1 and Ankyrins	4
NRAGE signals death through JNK	3
Pregnenolone biosynthesis	3

Rules firing most often in this group	HR	Patients
F5-R1: M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	0.32	11
F5-R2: M stage = M0 AND 26.5 ≤ Age < 83.5	0.35	11
F2-R8: 40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	0.47	9
F3-R7: M stage = M0 AND 40.5 ≤ Age < 86.5	0.37	9
F3-R2: N stage = N0	0.37	6
F2-R7: M stage = M0 AND 40.5 ≤ Age < 83.5	0.39	6

Most common driver alterations

PIK3CA mutated (29%), TP53 mutated (27%), CCND1 amplified (20%), MYC amplified (18%), GATA3 mutated (12%), FGFR1 amplified (12%), CDH1 mutated (12%), ERBB2 amplified (11%), MAP2K4 mutated (7%), CBFB mutated (7%)

Patient index

Patient	Age	Stage	Subtype	Recorded	Model estimate	Window	Risk pct	Report file
TCGA-B6-A0RI	44	III	LumA	36.2 mo	29 mo	✗	71	D_progression_after_36m/TCGA-B6-A0RI.pdf
TCGA-B6-A0X7	62	X	LumA	36.3 mo	24 mo	✗	92	D_progression_after_36m/TCGA-B6-A0X7.pdf
TCGA-B6-A0I1	73	II	Basal	36.8 mo	15 mo	✗	50	D_progression_after_36m/TCGA-B6-A0I1.pdf
TCGA-BH-A0C1	61	III	LumA	36.9 mo	29 mo	✗	77	D_progression_after_36m/TCGA-BH-A0C1.pdf
TCGA-A8-A08J	52	IV	Her2	37.1 mo	6 mo	✗	100	D_progression_after_36m/TCGA-A8-A08J.pdf
TCGA-AR-A1AH	51	II	Basal	37.3 mo	20 mo	✗	64	D_progression_after_36m/TCGA-AR-A1AH.pdf
TCGA-BH-A203	78	II	Her2	38.6 mo	30 mo	✗	66	D_progression_after_36m/TCGA-BH-A203.pdf
TCGA-E2-A15O	89	I	LumA	39.8 mo	8 mo	✗	1	D_progression_after_36m/TCGA-E2-A15O.pdf
TCGA-A2-A0SX	48	I	Basal	40.7 mo	23 mo	✗	24	D_progression_after_36m/TCGA-A2-A0SX.pdf
TCGA-A2-A25B	39	II	LumB	42.4 mo	38 mo	✓	13	D_progression_after_36m/TCGA-A2-A25B.pdf
TCGA-E2-A1LL	73	III	Basal	42.8 mo	17 mo	✗	93	D_progression_after_36m/TCGA-E2-A1LL.pdf
TCGA-AR-A2LK	62	III	LumB	42.9 mo	31 mo	✗	86	D_progression_after_36m/TCGA-AR-A2LK.pdf
TCGA-AR-A0TZ	43	III	LumA	43.2 mo	41 mo	✓	98	D_progression_after_36m/TCGA-AR-A0TZ.pdf
TCGA-B6-A0X5	61	II	LumB	44.2 mo	30 mo	✗	41	D_progression_after_36m/TCGA-B6-A0X5.pdf
TCGA-A2-A0EW	53	III	LumA	44.5 mo	32 mo	✗	77	D_progression_after_36m/TCGA-A2-A0EW.pdf
TCGA-A2-A0CK	60	III	LumA	44.8 mo	39 mo	✓	75	D_progression_after_36m/TCGA-A2-A0CK.pdf
TCGA-B6-A0IN	45	X	LumB	46.4 mo	24 mo	✗	91	D_progression_after_36m/TCGA-B6-A0IN.pdf
TCGA-AR-A24L	26	II	LumA	46.5 mo	51 mo	✓	38	D_progression_after_36m/TCGA-AR-A24L.pdf
TCGA-A2-A04V	39	II	LumA	48.3 mo	29 mo	✗	7	D_progression_after_36m/TCGA-A2-A04V.pdf
TCGA-BH-A1EX	67	II	LumA	49.6 mo	40 mo	✓	46	D_progression_after_36m/TCGA-BH-A1EX.pdf
TCGA-B6-A402	47	I	missing	49.8 mo	31 mo	✗	26	D_progression_after_36m/TCGA-B6-A402.pdf
TCGA-HN-A2OB	45	II	LumA	50.7 mo	62 mo	✓	22	D_progression_after_36m/TCGA-HN-A2OB.pdf
TCGA-AR-A0U2	47	III	LumB	52.9 mo	34 mo	✗	82	D_progression_after_36m/TCGA-AR-A0U2.pdf
TCGA-AO-A03U	31	I	Normal	53.1 mo	33 mo	✗	4	D_progression_after_36m/TCGA-AO-A03U.pdf
TCGA-E2-A1L7	40	III	LumB	53.2 mo	38 mo	✓	78	D_progression_after_36m/TCGA-E2-A1L7.pdf
TCGA-AR-A256	45	II	Basal	53.9 mo	16 mo	✗	63	D_progression_after_36m/TCGA-AR-A256.pdf
TCGA-BH-A1FR	73	III	LumA	54.0 mo	25 mo	✗	72	D_progression_after_36m/TCGA-BH-A1FR.pdf
TCGA-AO-A03N	59	II	LumB	54.1 mo	30 mo	✗	35	D_progression_after_36m/TCGA-AO-A03N.pdf
TCGA-BH-A1FL	69	II	LumA	55.0 mo	49 mo	✓	44	D_progression_after_36m/TCGA-BH-A1FL.pdf
TCGA-BH-A1FU	44	I	Normal	55.5 mo	32 mo	✗	6	D_progression_after_36m/TCGA-BH-A1FU.pdf
TCGA-BH-A1EW	38	II	LumA	55.7 mo	57 mo	✓	10	D_progression_after_36m/TCGA-BH-A1EW.pdf
TCGA-BH-A0BQ	39	I	LumA	56.2 mo	34 mo	✗	1	D_progression_after_36m/TCGA-BH-A0BQ.pdf
TCGA-3C-AAAU	55	X	LumA	59.4 mo	29 mo	✗	69	D_progression_after_36m/TCGA-3C-AAAU.pdf
TCGA-E2-A1IE	61	II	LumA	59.5 mo	35 mo	✗	21	D_progression_after_36m/TCGA-E2-A1IE.pdf
TCGA-AO-A03P	54	II	LumB	63.2 mo	35 mo	✗	29	D_progression_after_36m/TCGA-AO-A03P.pdf
TCGA-B6-A0WY	40	III	LumA	66.1 mo	20 mo	✗	77	D_progression_after_36m/TCGA-B6-A0WY.pdf
TCGA-B6-A0RS	38	II	Her2	67.1 mo	24 mo	✗	64	D_progression_after_36m/TCGA-B6-A0RS.pdf
TCGA-B6-A0RH	51	II	Her2	71.2 mo	21 mo	✗	71	D_progression_after_36m/TCGA-B6-A0RH.pdf
TCGA-BH-A1FN	34	II	LumB	72.1 mo	53 mo	✓	36	D_progression_after_36m/TCGA-BH-A1FN.pdf
TCGA-BH-A1FE	31	II	LumA	74.7 mo	69 mo	✓	10	D_progression_after_36m/TCGA-BH-A1FE.pdf
TCGA-A8-A08T	64	IV	LumA	84.1 mo	22 mo	✗	99	D_progression_after_36m/TCGA-A8-A08T.pdf
TCGA-AR-A24O	43	III	LumA	84.2 mo	27 mo	✗	82	D_progression_after_36m/TCGA-AR-A24O.pdf
TCGA-BH-A0C3	47	I	LumB	87.9 mo	36 mo	✗	29	D_progression_after_36m/TCGA-BH-A0C3.pdf
TCGA-B6-A0IB	64	IV	LumB	89.3 mo	7 mo	✗	99	D_progression_after_36m/TCGA-B6-A0IB.pdf
TCGA-BH-A18K	46	I	LumA	90.8 mo	34 mo	✗	2	D_progression_after_36m/TCGA-BH-A18K.pdf
TCGA-B6-A0RP	73	III	LumA	91.6 mo	15 mo	✗	90	D_progression_after_36m/TCGA-B6-A0RP.pdf
TCGA-A2-A25E	34	III	LumB	98.3 mo	41 mo	✓	76	D_progression_after_36m/TCGA-A2-A25E.pdf
TCGA-GM-A2DA	46	II	LumA	99.1 mo	39 mo	✓	15	D_progression_after_36m/TCGA-GM-A2DA.pdf
TCGA-B6-A0RU	40	I	Basal	101.1 mo	22 mo	✗	52	D_progression_after_36m/TCGA-B6-A0RU.pdf

Patient	Age	Stage	Subtype	Recorded	Model estimate	Window	Risk pct	Report file
TCGA-BH-A1ES	35	II	missing	113.8 mo	58 mo	✓	63	D_progression_after_36m/TCGA-BH-A1ES.pdf
TCGA-BH-A1FB	60	II	missing	120.6 mo	35 mo	✗	60	D_progression_after_36m/TCGA-BH-A1FB.pdf
TCGA-B6-A0WT	61	II	LumA	127.0 mo	30 mo	✗	17	D_progression_after_36m/TCGA-B6-A0WT.pdf
TCGA-AR-A2LE	69	I	LumA	138.2 mo	33 mo	✗	3	D_progression_after_36m/TCGA-AR-A2LE.pdf
TCGA-B6-A2IU	62	II	LumA	138.9 mo	31 mo	✗	7	D_progression_after_36m/TCGA-B6-A2IU.pdf
TCGA-B6-A0IG	50	II	LumA	146.5 mo	33 mo	✗	32	D_progression_after_36m/TCGA-B6-A0IG.pdf
TCGA-B6-A0WZ	50	II	LumA	168.2 mo	33 mo	✗	14	D_progression_after_36m/TCGA-B6-A0WZ.pdf